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Paleomagnetism and geochronology of late Paleozoic granitic rocks from the Lake District of southern Chile: Implications for accretionary tectonics

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ABSTRACT

K/Ar and $^{40}Ar/^{39}Ar$ dating shows that granitic rocks in the central Lake District, southern Chile, are late Paleozoic. Granitic rocks of similar age extend far up the coast of Chile, perhaps as much as 1800 km. We have obtained paleomagnetic directions from seven sites near Lago Ranco; these give a paleomagnetic pole at lat 57.4°S, long 323.5°E, with a circle of 95% confidence of 18.8°. This is not significantly different from the mean late Paleozoic reference pole for the South American craton. If our results are characteristic of the late Paleozoic belt as a whole, it follows that post–late Paleozoic accretion and terrane displacement are absent throughout much of the central Andes.

INTRODUCTION

Proof that accretion has played an essential role in the orogenic evolution of western North America has stimulated interest in whether the same is true for the Andes. Much of the paleomagnetic work done to date in Chile suggests that the central Andes did not experience important post-Triassic terrane accretion or along-coast displacement (e.g., Forsythe et al., 1987; Hartley et al., 1988; Irwin et al., 1987; Jesinkey et al., 1987; older literature reviewed by Beck, 1988). However, very little of the central Andes has been studied paleomagnetically, so the question has remained open. In this paper we present some results which suggest that post–late Paleozoic accretion is absent along much of the coastline of central Chile.

REGIONAL SETTING

Figure 1 shows the regional setting of the Lake District, which lies on the eastern side of the central valley of Chile. Most exposures are of granitic rocks of various ages (late Paleozoic to Miocene), young volcanic rocks, and Late Cretaceous and younger sedimentary rocks. K/Ar geochronology has shown that granitic rocks around Lago Ranco, previously thought to be Jurassic, are actually late Paleozoic (Table 1 and Munizaga et al., 1984, 1988). The rocks we sampled near Lagos Ranco and Riñihue are a variable suite of fresh-appearing, homogeneous, medium- to coarse-grained granitic rocks that range in composition from biotite granodiorite to pyroxene-bearing diorite. Probably more than one pluton is involved, but conditions of exposure and the state of geologic mapping do not let us say for sure. These rocks appear to be the southern end of an essentially continuous belt of late Paleozoic plutonic rocks that extends about 900 km northward along the coast to Valparaiso (Fig. 1) and then perhaps another 1000 km as sporadic exposures to near lat 23°S. South of Lago Ranco, late Paleozoic rocks seem to be absent, although a 310–320 Ma greenschist belt extends to lat 52°S (Halpern, 1973).

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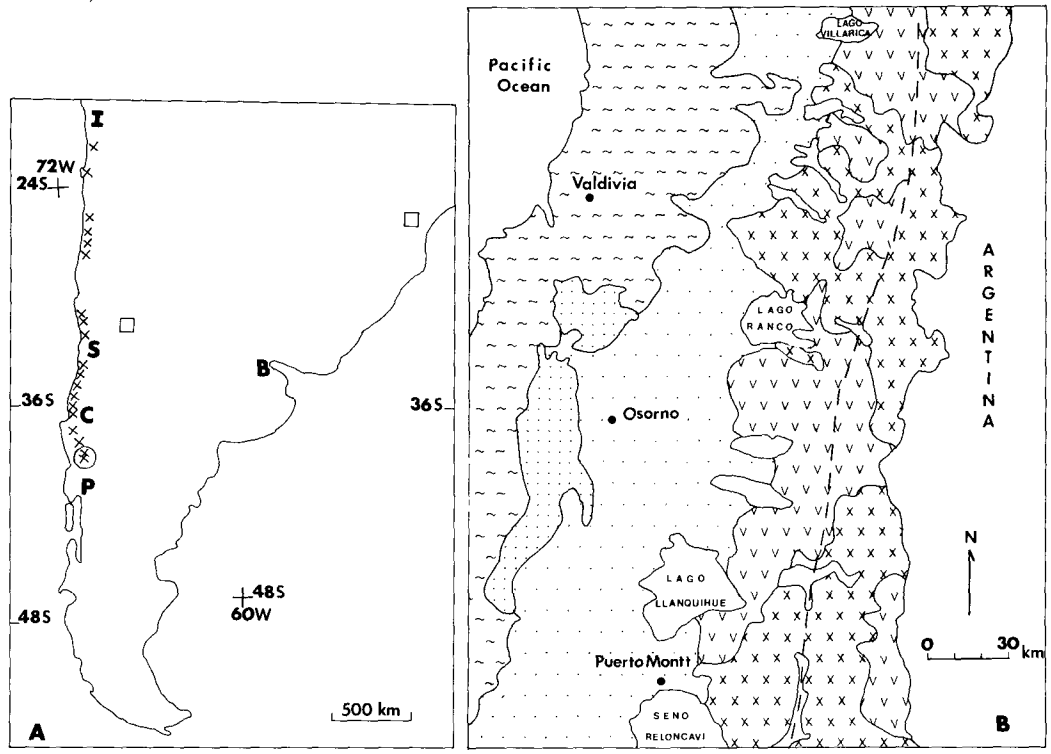


Figure 1. Location and regional geologic setting. A: Southern South America, with location of sampling area (circle). X pattern shows coastal late Paleozoic intrusive belt. P = Puerto Montt, C = Concepción, S = Santiago, I = Iquique, B = Buenos Aires. Squares show approximate location of sampling areas for reference poles. B: Geologic sketch map simplified from Munizaga et al. (1984). Patterns: X = granitic rocks, V = volcanic rocks, wavy lines = metamorphic rocks, close-spaced dots = Tertiary sedimentary rocks, sparse dots = alluvium. Dashed line = approximate trace of Liquiñe-Ofqui fault.

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A much-discussed feature in southern Chile is the somewhat mysterious Liquiñe-Ofqui fault (Thiele et al., 1986), shown in Figures 1B and 2. This feature, possibly 1000 km or more in length, is defined by a series of aligned fjords and river valleys cut into mainly young volcanic rocks in the north (Lake District and farther north) and into granitic rocks in the south. Horizontal mineral lineations indicative of strike-slip faulting are present in some localities but absent in others. Many observers regard the Liquiñe-Ofqui as a dextral strike-slip fault, of unknown displacement. It has been attributed to collision between South America and the Chile Rise by Forsythe and Nelson (1985) and to oblique subduction of the Nazca plate by Beck (1988). Our sampling area lies west of the fault, which means that our paleomagnetic results could help detect displacement of the area relative to the South American interior.

GEOCHRONOLOGY

Figure 2 shows K/Ar dates near Lago Ranco (Table 1 and Munizaga et al., 1984, 1988). Not shown in Figure 2 are dates from granitic rocks along the Liquiñe-Ofqui fault and farther east; these are much younger than the dates shown and range into the Pliocene. According to Munizaga et al. (1988), emplacement of Miocene plutons seems in part to have been controlled by the location of the fault.

One result of potential importance is from site 85cj39, which has normal magnetic polarity (Table 2); all other sites are reverse. As shown, 85cj39 has a K/Ar age of 309 ±8 Ma, Early Pennsylvanian (Bashkirian/Moscovian) according to the time scale of Harland et al. (1982). Thus it seems likely that these rocks were magnetized just before the beginning of the Kiaman (reverse polarity) magnetic interval (Irving and Parry, 1963). The beginning of the Kiaman is placed near the Bashkirian/Moscovian boundary by Khramov (1967), but seems to be younger in sections in Australia and North America (e.g., McElhinny, 1973, p. 125–128).

PALEOMAGNETISM

Results

Paleomagnetic samples were collected from the north and south shores of Lago Riñihue and around the eastern side of Lago Ranco (Fig. 2). Each site consisted of seven to nine samples spaced several metres apart at road cuts, quarries, or glaciated exposures. Samples were drilled and oriented in situ by using both sun and magnetic compass. In the laboratory, all samples were subjected to thermal and/or alternating field demagnetization, and characteristic directions of remanent magnetization were determined by using orthogonal diagrams and a standard line-fitting technique. Examples of laboratory demagnetization are shown in Figure 3.

Our results fall into two categories: good and terrible. In general, samples collected near Lago

TABLE 1. LASER FUSION BIOTITE AGES FROM GRANITIC ROCKS NEAR LAGO RIÑIHUE AND LAGO RANCO, SOUTHERN CHILE

Site								
P	G	$^{40}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{36}Ar/^{39}Ar$	$^{40}Ar^*/^{39}Ar$	$^{40}Ar^*$ (%)	J†	Age (Ma)
88bc1	RIN10	46.247	0.0676	0.00041	46.131	99.7	0.003862	297.05 ±1.44
88bc2	RIN11	46.405	0.0226	0.00792	44.064	95.0	0.003880	284.74 ±2.36
88bc3	RIN12	47.167	0.0454	0.00060	46.993	99.6	0.003880	307.16 ±2.27
88bc4	RIN13	46.999	0.0447	0.00023	46.934	99.9	0.003880	301.82 ±1.34
88bc5	RIN14	47.076	0.0250	0.00045	46.945	99.7	0.003880	301.88 ±1.63
88bc7	RAN15	50.64	0.435	0.00193	50.11	98.9	0.003673	304.81 ±1.15

Note: For sites, P is paleomagnetic and G is geochronological. For locations see Figure 2 and Table 2.

*Radiogenic ^{40}Ar.

†Neutron flux coefficient.

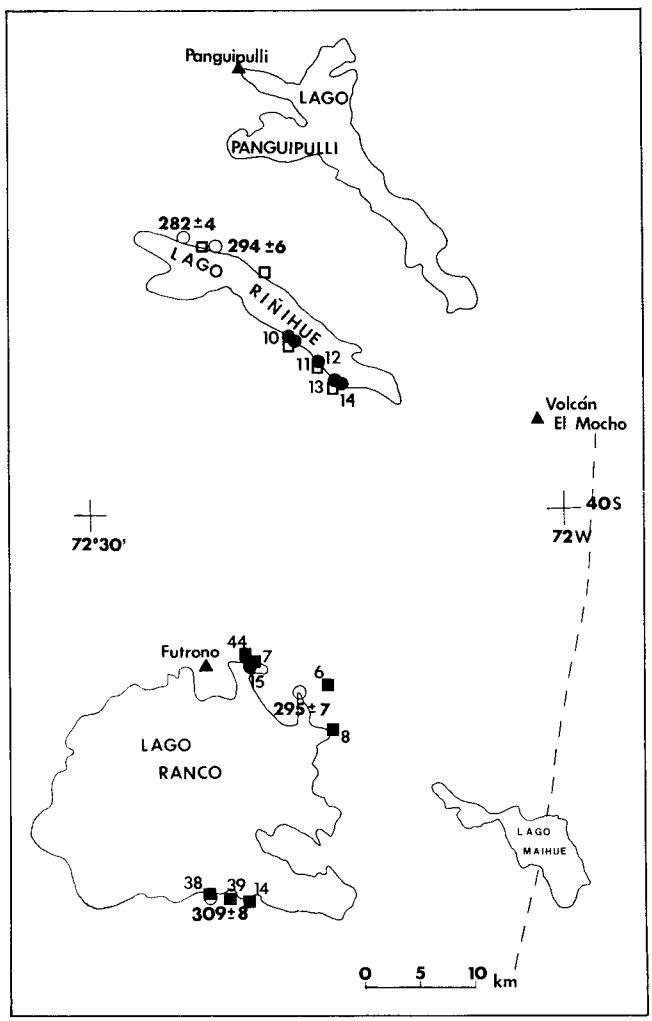


Figure 2. Geochronological (circles) and paleomagnetic (squares) sampling sites. Solid circles are new ages, from Table 1; open circles are previously published ages, from Munizaga et al. (1984, 1988). Solid squares are sites with useful paleomagnetic results (Table 2); open squares are sites for which no useful results were obtained. Approximate trace of Liquiñe-Ofqui fault shown.

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Ranco had errors in line fitting of < 15° (usually much less) and grouped to give site-mean directions with circles of 95% confidence (α_{95}) of about 10° or less—rarely up to 13°. Precision parameters for these sites ranged from about 15 to 450. In contrast, most samples from apparently identical rocks near Lago Riñinue had very noisy demagnetization paths characterized by errors in line fitting of >15° (usually much more). Moreover, samples that did yield reasonably well defined directions of characteristic remanent magnetization tended to group poorly, giving "clusters" with precision parameters approaching randomness and very large α_{95}. We have not determined why the Lago Riñihue samples behaved so badly, but we obviously need to exclude them from our pole calculations. Acceptable results (all from Lago Ranco) are presented in Table 2.

Reference Poles

Late Paleozoic reference poles for stable South America are reasonably well determined. Vilas (1981) listed six reference poles in the age range late Carboniferous–Early Permian; these are plotted in Figure 4A. The mean of these is lat 63.2°S, long 347.6°E, with α_{95} of 11.9° (Table 3). The mean late Paleozoic pole from Lago Ranco (lat 57.4°S, long 323.5°E) is 13.2° from this mean reference pole, not significant at 95% confidence. In view of the fact that the angular standard deviation of the South American reference poles also is 13.2°, our Lago Ranco result looks very much like a member of the distribution of late Paleozoic poles for stable South America.

INTERPRETATION

If the Lago Ranco pole is valid, it follows that the area has not moved much, relative to stable South America, since the close of the Paleozoic. Furthermore, if the Lago Ranco rocks form part of an unbroken late Paleozoic belt that extends along the Chilean coastline to beyond Valparaiso (Hervé et al., 1985), it also may follow that much of the central Chilean continental margin is similarly tied to the craton. This would imply that post-Paleozoic accretion has played little or no role in the orogenic development of the central Andes. Clearly, to establish this interpretation, we need several more poles, preferably some near the northern end of the belt (we have studied rocks from five such sites, but they are magnetically unstable). We also need to inquire further into the validity of the Lago Ranco pole.

Figure 4A shows the distribution of site-mean virtual geomagnetic poles (VGPs) from Lago Ranco, together with the mean late Paleozoic South American reference pole. It is clear from this illustration that the scatter of Lago Ranco VGPs is greater than for most paleomagnetic studies; for example, the angular standard deviation of Lago Ranco VGPs is 23.7°, compared to about 16°–20° for the same latitude for the past 110 m.y. (Merrill and McElhinny, 1983, Fig. 6.17). Only a small part of this scatter can be due to measurement error; other causes must be at work. We consider two.

Apparent Polar Wander

Figure 4A shows the trend of apparent polar wander (APW) for stable South America during the late Paleozoic, after Vilas (1981). If our sites were magnetized over a long interval (10 m.y. or more), we might expect the Lago Ranco data to

TABLE 2.
PALEOMAGNETIC RESULTS FROM PALEOZOIC GRANITIC ROCKS NEAR LAGO RANCO, SOUTHERN CHILE

Site	N	Dec (°)	Inc (°)	α_{95}	Site Lat (°S)	Site Long (°W)	VGP Lat (°S)	VGP Long (°E)
86cj38	7	110.4	70.7	3.8°	40.32	72.37	43.0	335.0
85cj39	8	343.4	-72.9	5.8	40.32	72.37	69.1	312.4
85cj44	8	136.5	69.8	2.3	40.12	72.33	58.0	338.0
88bc06	8	27.7	84.7	10.7	40.13	72.25	30.7	293.4
88bc07	8	189.3	78.6	9.5	40.12	72.33	61.6	280.4
88bc08	10	172.4	64.4	10.8	40.13	72.24	81.8	327.5
88bc14	9	110.0	50.2	13.2	40.32	72.36	33.8	3.5
Mean	7			18.8			57.4	323.5

Note: Site numbers keyed to Figure 2. N = number of specimens; Dec, Inc = declination, inclination; α_{95} = radius of circle of 95% confidence; VGP = virtual geomagnetic pole.

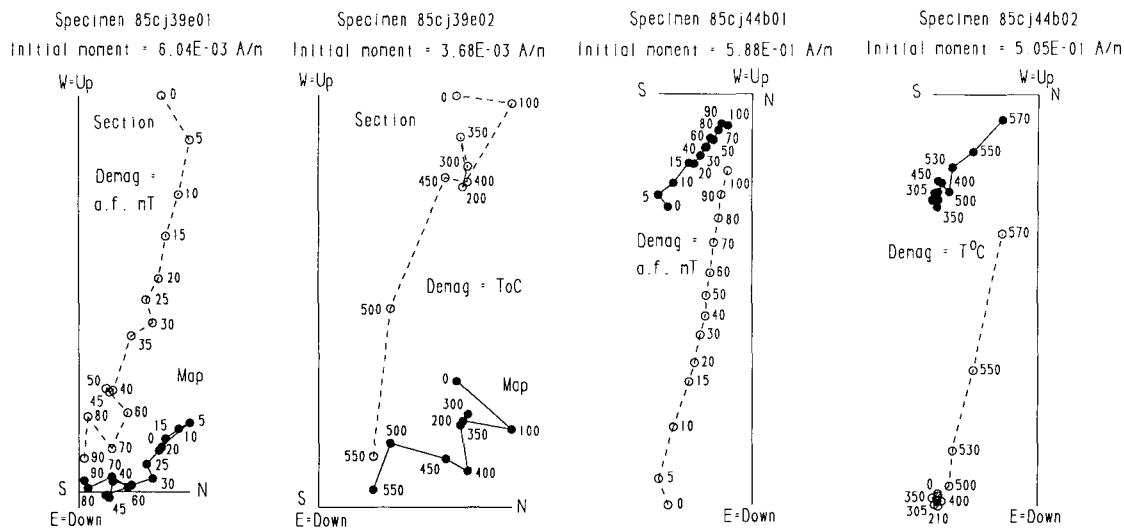


Figure 3. Examples of laboratory demagnetization by both alternating field and thermal methods; 85cj39 is normal polarity, and 85cj44 is reverse.

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be elongated in this direction. From Figure 4B there does seem to be a slight elongation in this direction, although not much. If the elongation is real and caused by APW, then sites 6, 14, and 38 ought to be considerably older than 7 and 8. To some extent this is confirmed by the K/Ar dating; the northern side of Lago Ranco (paleomagnetic sites 6, 7, 8, and 44) appears to be younger than its southern side (sites 14, 38, and 39) (Fig. 2). If the two groups of sites are considered separately, the means are 24° apart and lie near the APW path of Vilas (1981): S group: lat 50.2°S, long 343.2°E, α_{95} 31.6°; N group: lat 60.2°S, long 304.1°E, α_{95} 26.1°. However, the confidence limits on these calculations are so large that it is difficult to estimate how much of the scatter in Figure 4B may be due to APW.

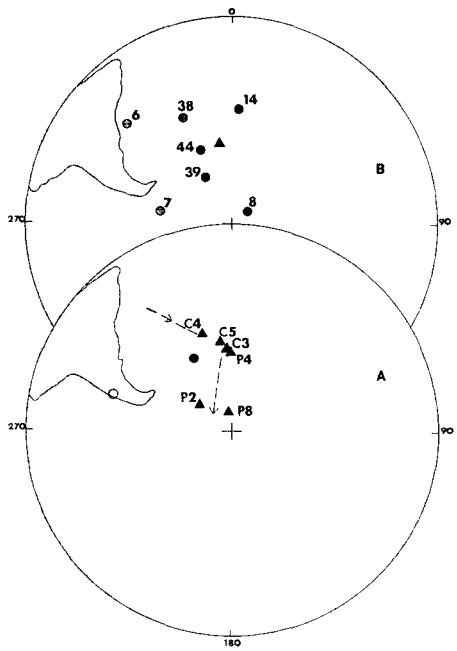


Figure 4. Paleomagnetic results. A: Reference poles, from Vilas (1981). Solid circle is mean pole from Lago Ranco. Open circle shows Lago Ranco area. B: Lago Ranco VGP; numbers are abbreviated site numbers from Table 2. Triangle is mean reference pole.

TABLE 3. LATE PALEOZOIC SOUTH AMERICAN REFERENCE POLES

Pole	Lat (°S)	Long (°E)
SAP8	82	349
SAP2	74	308
SAP4	59	359
SAC3	57	357
SAC5	54	351
Mean (all)	63.2	347.6 (α_{95}, 11.9°)
Mean (oldest 4)	54.9	351.9 (α_{95}, 6.8°)

Note: Pole designations after Vilas (1981); α_{95} is a radius of circle of 95% confidence.

Tilt

All seven Lago Ranco sites are from granitic rocks with no internal structure, so it is impossible to correct their paleomagnetic directions for postmagnetization tilt. Nearly flat lying Triassic sedimentary rocks crop out just west of Lago Riñihue, suggesting that that area has not been strongly deformed. However, there are small outcrops of younger (probably Late Cretaceous) sedimentary beds east and south of Lago Ranco that have variable tilts ranging up to about 45°. Strikes are north and northeast, with dips both to the east and west. Thus, blocks within the region may have undergone some tilt, although there is no evidence for *systematic* tilt such as might make a major shift in the mean direction and VGP. However, much of the "excess" scatter seen in Figure 4B may be caused by uncorrected tilt.

CONCLUSIONS

Figure 5 shows mean late Paleozoic poles for the Lago Ranco region and for the South American craton. The two are indistinguishable at 95% confidence, suggesting that the Lago Ranco region has not moved relative to the South American craton since late Paleozoic time. Because the Lago Ranco region appears to be the southern end of an essentially unbroken belt of late Paleozoic rocks that extends up the coast far to the north (Fig. 1), our results also suggest that no important post-Paleozoic allochthonous terranes exist in much of the central Andes. The paleomagnetic evidence would be much stronger if we had pinned the northern end of the belt to South America with another paleomagnetic study, but so far we have not been able to do so. However, directions reported in late Paleozoic rocks from northern Chile that are not part of the belt (Jesinkey et al., 1987) tend to support the hypothesis that the Chilean late Paleozoic rocks are autochthonous.

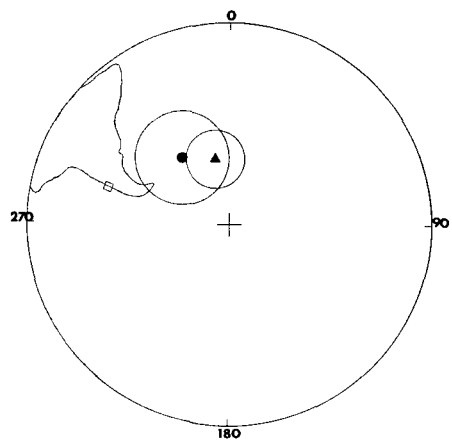


Figure 5. Mean late Paleozoic reference pole for South America (triangle) and mean Lago Ranco late Paleozoic pole (circle), with 95% confidence circles. Square shows Lago Ranco area.

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